SPECTRA — Project Architecture & Module 4 Deep-Dive

Project: SPECTRA (AI-Driven PQC Reverse Proxy for Zero-Trust) | **Lead Module:** M4 — AI Risk Engine | **Dataset:** UNSW-NB15 Benchmark | **Status:** Phase 1 Final Defense Ready

> **Document Purpose:** Comprehensive technical guide and viva defense reference for SPECTRA and Module 4 (AI Risk Engine). Every metric, threshold, architectural boundary, and code implementation detail is verified against the finalized codebase.

1. System Abstract & The Big Picture

The Problem in Traditional Networks

Modern enterprise networks face two distinct existential security threats:

1. **The Quantum Threat ("Harvest Now, Decrypt Later")**: Traditional TLS relies on RSA and ECC. Adversaries are actively capturing and archiving encrypted enterprise network traffic today. Once a Cryptographically Relevant Quantum Computer (CRQC) emerges, Shor's algorithm will crack RSA and ECC retrospectively, decrypting years of historical corporate secrets.
2. **Broken Perimeters & Lateral Movement**: Traditional networks authenticate users once at the perimeter (VPN). Once inside, the network implicitly trusts the session. If an attacker steals a token or compromises an endpoint, they can move laterally across internal databases undetected.

What SPECTRA Does

SPECTRA (**S**ecure **P**ost-quantum **E**nforcement & **C**ontinuous **T**elemetry **R**isk **A**nalysis) is a Zero-Trust Reverse Proxy that terminates client connections at the network edge in front of protected backend microservices.

- **Transport Security (M1)**: Enforces post-quantum lattice cryptography (NIST ML-KEM and ML-DSA) with hardware-aware dynamic parameter scaling.
- **Edge Routing & Enforcement (M2)**: Intercepts every individual request on port `8443`, routing verified requests to backend port `9000`.

- **Telemetry Extraction (M3)**: Leverages Linux in-kernel **eBPF probes** to extract non-intrusive flow statistics without user-space buffer copies.
- **Intelligence Plane (M4)**: Evaluates flow telemetry using an unsupervised **Isolation Forest** with **empirical normal percentile calibration**, outputting continuous risk scores in `[0.0, 1.0]` within 5.7 milliseconds.



2. Individual Module Overviews (M1, M2, M3)

Module	Core Technology	Responsibility	Current Status
M1	NIST PQC (ML-KEM, ML-DSA) via C/C++ `liboqs`	Replaces classical TLS with post-quantum key encapsulation and signatures. Dynamically downshifts parameter sets (Level 5 $\rightarrow$ Level 3 $\rightarrow$ Level 1) to accommodate resource-constrained IoT devices without packet fragmentation.	**IMPLEMENTED** Standalone benchmarking & parameter downshifting proven.
M2	Reverse Proxy Core in Go	Terminates public traffic at edge port `8443`. Isolates backend port `9000`. Coordinates verification pipeline, checks risk scores against thresholds, and executes ALLOW, CHALLENGE, or DROP policies.	**IMPLEMENTED** Standalone Go proxy functional; currently uses a `0.12` risk stub in Phase 1 testing.
M3	eBPF Kernel Probes & Suricata IDS	Attaches sandboxed eBPF probes directly into kernel socket hooks to measure packet lengths, rates, and inter-arrival intervals without CPU copy overhead. Integrates Suricata for signature-based CVE matching.	**IMPLEMENTED** Standalone eBPF sniffer & signature matching verified.

3. Deep-Dive: Module 4 (AI Risk Engine)

Why Unsupervised Anomaly Detection?

Traditional signature systems fail against zero-day exploits. Supervised machine learning models fail in network security due to the **labeling bottleneck** (zero-day attacks cannot be labeled in advance). M4 is strictly **unsupervised**: it is trained **exclusively on normal baseline traffic (`label == 0`)**. It learns the mathematical density of benign traffic so that any abnormal outlier is immediately flagged.

The 4 Network-Flow Features

M4 represents each network observation as a 4-dimensional continuous vector:

```
x = [packet_size, request_rate, connection_count, inter_arrival_time]
```

Feature Name	UNSW-NB15 Column	Physical Meaning	Normal Example	Anomalous Example
`packet_size`	`smean`	Mean byte size per transmitted packet.	`520 B` (Standard HTTP GET)	`9,500 B` (Buffer overflow / data exfil)
`request_rate`	`rate`	Transmitted requests / packets per second.	`6.0 req/s` (Human browsing)	`300.0 req/s` (Automated flood / fuzzer)
`connection_count`	`ct_srv_src`	Concurrent active sockets to same service.	`3.0` (Normal browser sockets)	`80.0` (Reconnaissance / DoS probe)
`inter_arrival_time`	`sinpkt / 1000.0`	Time interval between consecutive packets (sec).	`0.180 s` (Human pacing)	`0.002 s` (Automated script saturation)

Isolation Forest Mechanics

- **Core Principle**: "Anomalies are few and different." Outliers in high-dimensional space require far fewer random splits to isolate than normal points packed in dense clusters.

- **Tree Construction**: Builds an ensemble of 100 binary trees (`n_estimators = 100`). Each tree recursively partitions feature space by randomly choosing a feature and split value.
- **Path Length to Score**:
  - Dense normal points $\rightarrow$ Deep path length (Depth 10–14) $\rightarrow$ `decision_function > 0` .
  - Sparse anomalies $\rightarrow$ Shallow path length (Depth 1–4) $\rightarrow$ `decision_function < 0` .
- **Efficiency**: Linear time complexity $O(n)$ with inference execution in **~5.7 ms**.

Empirical Percentile Calibration

Rather than using arbitrary sigmoid normalization ($1 / (1 + e^{-k \cdot x})$), M4 calibrates raw scores against **11,200 held-out normal samples** that the model never saw during tree fitting.

1. In training (`src/train_model.py`), anomaly scores of held-out normal traffic are calculated and sorted: `anomaly_scores = -model.decision_function(X_calibration)`
2. In inference (`src/m4_interface.py`), the incoming score is ranked via binary search: `position = np.searchsorted(REFERENCE_ANOMALY_SCORES, anomaly_score, side="right")`
3. The percentile risk is: `Risk = position / 11200 ∈ [0.0, 1.0]`

> **Precise Meaning of Risk = 0.970:**
> A risk score of 0.970 means: "*This telemetry vector is more anomalous than approximately 97.0% of verified normal baseline traffic.*" It is an empirical percentile rank, not an attack probability.

Calibrated Zero-Trust Policy Thresholds

- **ALLOW (Risk < 0.970)**: Normal transaction. Injects `X-Spectra-Verified` header and forwards to backend.
- **MFA / MONITOR (0.970 ≤ Risk < 0.995)**: Elevated anomaly. Triggers step-up authentication. Setting this at 0.970 mathematically guarantees that at most ~3% of legitimate normal traffic will ever be challenged.
- **DROP (Risk ≥ 0.995)**: Extreme anomaly. Corresponds to the upper 0.5% of the calibrated normal reference distribution. Severs TCP connection immediately with HTTP 403.

Risk Score	Raw Decision Score	Empirical Distribution Position	Enforced Policy Action
0.070	`+0.1546`	Deep in normal cluster (7th percentile).	**ALLOW**
0.850	`+0.0410`	Normal operational variance (85th percentile).	**ALLOW**
0.980	`-0.0150`	Elevated deviation; exceeds 98% of normal baseline.	**MFA / MONITOR**
1.000	`-0.2144`	Exceeds 100% of normal baseline calibration samples.	**DROP**

4. End-to-End Walkthrough: 4 Real-World Scenarios

Scenario 1: Normal Employee Accessing Internal Application

- **Input**: `packet_size = 520 B`, `request_rate = 6.0 req/s`, `connection_count = 3.0`, `inter_arrival_time = 0.180 s`
- **M1 Transport**: Client negotiates ML-KEM-768 / ML-DSA-65 session.
- **M2 Proxy**: Intercepts `GET /dashboard` on port 8443.
- **M3 Telemetry**: eBPF measures standard HTTP header packet size and steady human pacing.
- **M4 Vector**: `x = [520.0, 6.0, 3.0, 0.180]`
- **Isolation Forest**: Traverses deep tree path (Depth ~11.2). Raw score: `+0.1546`.
- **Percentile Risk**: Position 784 / 11,200 $\rightarrow$ **Risk Score = 0.070**.
- **Policy Gate**: `0.070 < 0.970` $\rightarrow$ **ALLOW**. Injects `X-Spectra-Verified: true`, forwards to backend:9000. HTTP 200 OK returned.

Scenario 2: Employee Unusually Rapid Activity (Moderate Anomaly)

- **Input**: `packet_size = 523 B`, `request_rate = 8.4 req/s`, `connection_count = 4.8`, `inter_arrival_time = 0.143 s`
- **M1 Transport**: Valid quantum-safe session established.
- **M2 Proxy**: User clicks rapidly or an automated browser tab reloader executes.
- **M3 Telemetry**: eBPF detects inter-arrival intervals shortening and frequency rising above normal human baseline.
- **M4 Vector**: `x = [523.0, 8.4, 4.8, 0.143]`
- **Isolation Forest**: Tree paths shorten (Depth ~7.8). Raw score: `-0.0150`.
- **Percentile Risk**: Position 10,976 / 11,200 $\rightarrow$ **Risk Score = 0.980**.
- **Policy Gate**: `0.970 ≤ 0.980 < 0.995` $\rightarrow$ **MFA / MONITOR**. Proxy holds session and issues WebAuthn/TOTP challenge.

Scenario 3: Attacker Making Rapid Probing Connections (Reconnaissance)

- **Input**: `packet_size = 64 B`, `request_rate = 45.0 req/s`, `connection_count = 38.0`, `inter_arrival_time = 0.022 s`
- **M1 Transport**: Client opens multiple concurrent TLS handshakes.
- **M2 Proxy**: Receives rapid connection attempts targeting varied microservice endpoints.
- **M3 Telemetry**: eBPF socket map detects spike in concurrent sockets (`ct_srv_src=38`) and minimal packet sizes.
- **M4 Vector**: `x = [64.0, 45.0, 38.0, 0.022]`
- **Isolation Forest**: Trees isolate observation quickly (Depth ~4.2). Raw score: `-0.1680`.
- **Percentile Risk**: Position 11,180 / 11,200 $\rightarrow$ **Risk Score = 0.998**.
- **Policy Gate**: `0.998 ≥ 0.995` $\rightarrow$ **DROP**. M2 severs TCP socket immediately. Returns HTTP 403 Forbidden.

Scenario 4: Volumetric Attack / Buffer Exploit (Obvious Malicious Attack)

- **Input**: `packet_size = 9500 B`, `request_rate = 300.0 req/s`, `connection_count = 80.0`, `inter_arrival_time = 0.002 s`
- **M1 Transport**: Massive inbound packet burst saturates buffer.
- **M2 Proxy**: High socket contention on edge port 8443.
- **M3 Telemetry**: eBPF detects maximum-MTU packet bursts and zero inter-arrival pauses (continuous saturation).
- **M4 Vector**: `x = [9500.0, 300.0, 80.0, 0.002]`
- **Isolation Forest**: Trees isolate observation on the first or second split (Depth ~2.1). Raw score: `-0.2144`.
- **Percentile Risk**: Anomaly score exceeds maximum reference calibration score $\rightarrow$ **Risk Score = 1.000**.
- **Policy Gate**: `1.000 ≥ 0.995` $\rightarrow$ **DROP**. Immediate TCP RST packet transmitted. IP temporarily blacklisted.

5. Official UNSW-NB15 Benchmark Evaluation

Evaluated on the official UNSW-NB15 test set containing **82,332 test samples** (37,000 normal, 45,332 real attack flows):

Metric	Measured Result	Engineering & Academic Explanation
Precision	**83.12%** (`0.8312`)	High alert reliability: When M4 raises an alarm ($\text{Risk} \ge 0.970$), it is an actual attack ~83% of the time.
Recall	**17.78%** (`0.1778`)	Catches 8,060 out of 45,332 attacks. Lower recall is expected because 4 flow features cannot detect single-packet stealth exploits.
F1-Score	**29.29%** (`0.2929`)	Harmonic mean balancing precision and recall ($2 \cdot P \cdot R / (P + R)$); pulled down by the prototype's lower recall.
False-Positive Rate	**4.42%** (`0.0442`)	Out of 37,000 normal test flows, only 1,637 triggered challenges, closely matching the theoretical ~3% design target.
ROC-AUC	**0.7570**	Strong continuous ranking capability across all potential decision thresholds.
Median Latency	**5.720 ms**	Single-pass inference per request, well under enterprise proxy SLA limits.
P95 Latency	**8.439 ms**	95% of all inferences complete within 8.5 ms.

Confusion Matrix (N = 82,332)

	Predicted Normal (Risk < 0.970)	Predicted Attack (Risk ≥ 0.970)
Actual Normal (37,000)	**TN: 35,363** (95.58%)	**FP: 1,637** (4.42%)
Actual Attack (45,332)	**FN: 37,272** (82.22%)	**TP: 8,060** (17.78%)

Attack Category Recall Breakdown

Attack Category	Recall	Technical Analysis (Fact vs. Inference)
Generic	**37.78%**	*Verified Fact:* High-volume script attacks create massive deviations in `rate` and `connection_count`.
Fuzzers	**8.43%**	*Technical Inference:* Random boundary-testing payload mutations cause detectable outliers in `packet_size`.
Backdoors	**5.32%**	*Technical Inference:* Persistent reverse shells slightly alter keep-alive intervals and `connection_count`.
Analysis	**4.87%**	*Technical Inference:* Port sweeps and scanning tools alter connection frequencies and timing.
Worms	**4.55%**	*Technical Inference:* Automated self-propagation patterns occasionally create slight volumetric anomalies.
Shellcode	**4.50%**	*Technical Inference:* Small executable payloads occasionally cause mild packet size variance.
DoS	**2.74%**	*Verified Observation:* In UNSW-NB15, many DoS traces use spaced, distributed probing that evades single-source rate counters.
Reconnaissance	**1.60%**	*Verified Fact:* Stealth probes use standard packet sizes and slow timing, evading flow-only models.
Exploits	**1.52%**	*Verified Fact:* Single-packet payload exploits (SQLi, buffer overflows) travel inside normal HTTP packets. Blind without IDS payload inspection.

> **Synthetic vs. Real Evaluation:**
> Synthetic testing (`evaluate_model.py`) showed **100% recall** because synthetic anomalies were artificially generated with massive spikes (10,000B packets, 300 req/s). Real-world UNSW-NB15 traffic showed **17.78% recall** because real attacks deliberately blend into normal traffic. *Never present synthetic testing as proof of real detection accuracy.*

6. Current Architecture Reality: Implemented vs. Planned

Component / Claim	Classification	Exact Technical Reality in Codebase
M1 Adaptive PQC	**IMPLEMENTED**	Standalone C/C++ `liboqs` engine; verified dynamic Level 5 to Level 3 downshifting based on CPU cores.
M2 Zero-Trust Reverse Proxy	**IMPLEMENTED**	Standalone Go reverse proxy on port 8443 routing to port 9000; enforces rules and header injection.
M3 eBPF Telemetry Probe	**IMPLEMENTED**	Standalone in-kernel eBPF socket monitoring and Suricata IDS signature checks.
M4 Isolation Forest Engine	**IMPLEMENTED**	Unsupervised Isolation Forest on 4 flow features with empirical percentile calibration.
UNSW-NB15 Benchmark Evaluation	**EVALUATED**	Evaluated on 82,332 test samples; precision (83.12%), ROC-AUC (0.7570), and latency (5.72 ms) verified.
Live M2 $\rightarrow$ M4 Dynamic HTTP Link	**PLANNED**	In Phase 1 standalone testing, M2 uses a static risk score stub of `0.12`. Dynamic HTTP querying planned for Phase 2.
Live M3 $\rightarrow$ M4 Feature Window Streaming	**PLANNED**	Live streaming of aggregated feature windows from eBPF ring buffers into M4 planned for Phase 2.
Multi-Modal Feature Fusion	**PLANNED**	Combining session identity, geolocation, and IDS deep-packet signatures into M4 planned for Phase 2.

7. Master Viva Defense Preparation

The Elevator Pitches

30-Second Elevator Pitch (The Entire Project)

"SPECTRA is an AI-driven, Post-Quantum Zero-Trust Reverse Proxy designed to secure internal microservices against both future quantum decryption and current lateral network movement. At the transport layer, it implements hardware-adaptive NIST PQC primitives (ML-KEM and ML-DSA). At the policy layer, it continuously scores network-flow telemetry using an unsupervised Isolation Forest calibrated with empirical percentiles, dynamically enforcing step-up MFA or connection dropping with sub-10ms inference latency."

1-Minute Pitch (The Entire Project)

"Traditional perimeter security suffers from two critical flaws: legacy public-key encryption like RSA and ECC is vulnerable to 'Harvest Now, Decrypt Later' quantum attacks, and static single-login perimeters fail once an adversary bypasses the boundary. SPECTRA addresses both. Incoming connections negotiate NIST-standardized quantum-safe algorithms—ML-KEM for key encapsulation and ML-DSA for digital signatures—adapting dynamically to client compute capacity. Behind this transport layer, an edge reverse proxy coordinates with eBPF kernel probes to extract flow telemetry. This telemetry is evaluated by an unsupervised Isolation Forest that maps flow vectors against empirical normal percentiles. Sessions with normal risk are permitted, moderate deviations trigger MFA challenges, and extreme anomalies are severed at the edge—delivering continuous verification with minimal latency."

2-Minute Pitch (Module 4 Deep-Dive — Your Module)

"I am responsible for Module 4: the AI Risk Engine. Traditional intrusion detection relies on static signatures or supervised models, which fail against novel zero-day attacks and suffer from severe labeling bottlenecks. In M4, I implemented an unsupervised anomaly detection engine powered by an Isolation Forest, trained exclusively on normal network traffic across four core flow features: packet size, request rate, connection count, and inter-arrival time. Rather than applying arbitrary sigmoid transformations, I engineered an empirical percentile calibration mechanism. We train the model on 80% of normal traffic and calibrate it against a held-out 20% normal baseline. When a live telemetry vector is evaluated, M4 calculates its exact percentile rank against this held-out distribution. This enables mathematically grounded zero-trust policy thresholds: traffic below the 97th percentile is allowed; traffic between the 97th and 99.5th percentiles triggers step-up MFA; and extreme outliers in the top 0.5% are dropped. I benchmarked M4 against the official 82,332-sample UNSW-NB15 dataset. The engine achieved 83.12% precision and a 4.42% false-positive rate, with a median inference latency of 5.72 ms. While volumetric attacks were detected effectively, stealthy payload exploits confirmed that pure flow features alone cannot catch single-packet attacks—providing empirical justification for our Phase 2 multi-modal telemetry fusion."

Top 10 High-Yield Viva Questions & Answers

Q1: What does "unsupervised" mean in M4, and how is it trained?

Answer: The model is trained **strictly on normal baseline traffic (** `label == 0` **)**. It is never fed attack signatures or labels during training. It learns what normal traffic looks like so that any statistical outlier is flagged as an anomaly.

> **Examiner follow-up:** *"What is the contamination parameter for?"* $\rightarrow$ *"In scikit-learn,* `contamination=0.05` *sets the offset for the raw decision boundary, assuming ~5% of baseline network records contain mild noise."*

Q2: How do you convert raw Isolation Forest scores into a 0 to 1 risk score?

Answer: We negate `model.decision_function(X)` so higher values represent greater anomalies, then execute `np.searchsorted()` against our sorted array of 11,200 held-out normal anomaly scores. The index divided by 11,200 gives the empirical percentile risk score in $[0.0, 1.0]$.

> **Examiner follow-up:** *"What is the physical meaning of a 0.970 score?"* $\rightarrow$ *"It means the observation is more anomalous than 97.0% of verified normal traffic."*

Q3: Why are your thresholds set to 0.970 and 0.995 instead of 0.50 or 0.70?

Answer: Because in an empirical percentile model, normal traffic is uniformly distributed across $[0.0, 1.0]$. A 0.50 threshold would challenge 50% of legitimate users. Setting MFA at 0.970 mathematically bounds false positive challenges on normal traffic to ~3%. Setting DROP at 0.995 reserves termination strictly for the upper 0.5% extreme anomalies.

Q4: Why is your recall only 17.78% on UNSW-NB15? Is that acceptable?

Answer (CRITICAL HONEST ANSWER): Recall is 17.78% because the prototype analyzes **only 4 network-flow features**. Single-packet exploits (SQLi, buffer overflows) travel in standard-sized packets at normal speed, mimicking normal flow headers. While low, this prototype successfully catches volumetric and flood attacks (Generic at 37.8%) with zero signatures, and empirically proves why Phase 2's cross-layer fusion with Suricata IDS signatures is required.

Q5: What does an ROC-AUC of 0.7570 indicate?

Answer: It proves solid continuous discriminative power. If you randomly select one attack record and one normal record from the 82,332 test samples, the model will correctly rank the attack record as more anomalous 75.7% of the time.

Q6: Why did you split normal data into training and calibration sets?

Answer: To avoid overfitting. Calibrating percentiles on the same data used to construct the isolation trees would yield overconfident percentiles. Using a held-out 20% normal split guarantees unbiased calibration.

Q7: Why is M4 inference latency (~5.7 ms) so critical?

Answer: M4 sits in the inline evaluation path of a reverse proxy. If inference took 150 ms, every page asset would suffer extreme delays. A 5.7 ms median latency fits comfortably within modern zero-trust SLA budgets.

Q8: Is M4 currently integrated with M2 in live real-time?

Answer (TRAP QUESTION): In the current Phase 1 codebase, M2 and M4 are validated as **standalone modules**. M2 currently uses a static risk score stub of `0.12` during proxy execution. Live dynamic HTTP/gRPC querying between M2 and M4 is designed and planned for Phase 2.

Q9: Why use an Isolation Forest instead of One-Class SVM or Autoencoders?

Answer: Isolation Forest exhibits linear time complexity $O(n)$, low memory requirements, and sub-10ms inference. One-Class SVM scales quadratically ($O(n^2)$) and Autoencoders require GPU hardware and intensive tuning that introduce latency.

Q10: How does SPECTRA mitigate the "Harvest Now, Decrypt Later" threat?

Answer: By using NIST-standardized Post-Quantum Cryptography (ML-KEM and ML-DSA) at the transport layer, ensuring that intercepted ciphertext cannot be decrypted by Shor's algorithm on a quantum computer.

8. One-Page Quick Revision Cheat Sheet

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================================================================================
                            SPECTRA PROJECT CHEAT SHEET (VIVA REVISION)
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PROJECT TITLE:
SPECTRA – AI-Driven PQC Reverse Proxy for Zero-Trust

CORE PURPOSE:
A Zero-Trust reverse proxy sitting in front of internal microservices that stops post-
decryption at the transport layer (M1) and prevents lateral movement at the edge via c
AI behavioral telemetry risk scoring (M4).

THE 4 MODULES:
• M1 (Adaptive PQC Transport): NIST FIPS 203 (ML-KEM) & FIPS 204 (ML-DSA) via liboqs (
  Adaptive parameter sets downshift based on client compute (Level 5 / Level 3 / Level
• M2 (Zero-Trust Proxy Engine): Go reverse proxy intercepting edge traffic on 8443, fo
  internal backend 9000. Re-verifies every request. Enforces ALLOW / CHALLENGE / DROP.
• M3 (IDS & Telemetry Extraction): Kernel-level eBPF probes capturing flow metadata wi
  Suricata deep-packet inspection for CVE signatures.
• M4 (AI Risk Engine): Unsupervised Isolation Forest evaluating 4 flow features, calib
  empirical normal percentiles to output low-latency risk scores in [0.0, 1.0].

M4 FEATURES (4 Network-Flow Dimensions):
1. packet_size (UNSW: smean)          -> Mean bytes per packet transmitted by source.
2. request_rate (UNSW: rate)          -> Transmission rate in requests/packets per sec
3. connection_count (UNSW: ct_srv_src)-> Concurrent active sockets to same service.
4. inter_arrival_time (UNSW: sinpkt)  -> Time elapsed between consecutive packets (sir

ISOLATION FOREST MECHANICS:
• Unsupervised ensemble of 100 random decision trees.
• Core principle: "Anomalies are few and different." Outliers are isolated near the ro
• Trained strictly on NORMAL traffic (label == 0). Never sees attack labels during tra

EMPIRICAL PERCENTILE CALIBRATION:
• Normal training data split into: 80% Fit (44,800 rows) and 20% Held-Out Calibration
• Raw score = -decision_function(x).
• Risk score = empirical rank in held-out normal calibration array: risk = rank / 11,2
• Interpretation: "0.970 = more anomalous than 97.0% of held-out normal traffic."

ZERO-TRUST POLICY THRESHOLDS:
• Risk < 0.970              -> ALLOW (normal access, injects X-Spectra-Verified header
• 0.970 <= Risk < 0.995     -> MFA / MONITOR (challenges user with step-up verificatio
• Risk >= 0.995             -> DROP (top 0.5% extreme anomaly; severs TCP connection w

OFFICIAL UNSW-NB15 BENCHMARK EVALUATION (82,332 Test Samples):
• Normal test samples : 37,000      • Attack test samples : 45,332
• Precision           : 83.12%      • False Positive Rate : 4.42%
• Recall              : 17.78%      • ROC-AUC             : 0.7570
• F1-Score            : 29.29%      • Latency (Median/P95): 5.720 ms / 8.439 ms
```

```
• Confusion Matrix    : TN = 35,363 | FP = 1,637 | FN = 37,272 | TP = 8,060

ATTACK CATEGORY RECALL BREAKDOWN:
• Generic: 37.78%  • Fuzzers: 8.43%  • Backdoors: 5.32%  • Analysis: 4.87%  • Worms: 4
• Shellcode: 4.50% • DoS: 2.74%      • Recon: 1.60%      • Exploits: 1.52%

KEY TAKEAWAY ON RESULTS:
• High precision (83.12%) and low FPR (4.42%) prove the reliability of percentile cali
• Low recall (17.78%) is honest: single-packet stealth exploits mimic flow headers. Th
  justifies Phase 2's cross-layer fusion with Suricata IDS signatures and identity cor

STATUS DISTINCTIONS (NEVER CLAIM PLANNED AS IMPLEMENTED):
• IMPLEMENTED : Standalone M1 PQC, Standalone M2 Go proxy, Standalone M3 eBPF, Standal
• EVALUATED   : M4 on UNSW-NB15 (82,332 rows); M1 downshifting latency; M4 inference l
• PLANNED     : Live M2 -> M4 HTTP dynamic query (M2 currently uses 0.12 stub); live M
                Multi-modal fusion (Identity, Geolocation, Device, IDS Deep Signatures
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